

Genomic Epidemiology of Highly Pathogenic Avian Influenza Virus H5N1 in Wild Birds in South Korea during 2021-2022; Changes in Viral epidemic patterns

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Introduction

- The major outbreak of high pathogenic avian influenza (HPAI) predominantly manifested in poultry, although sporadic spill-over events to wild birds were observed.
- However, the clade 2.3.4.4 A/goose/Guangdong/1/96 (Gs/GD) lineage HPAI viruses have continuously caused outbreaks among wild birds worldwide.
- As infection persists in wild birds, H5N8 HPAI clade 2.3.4.4b have reassorted with Low pathogenic avian influenza (LPAI) resulting in several subtypes, N1, N3, N4, N5, and N6.
- Clade 2.3.4.4b caused four outbreaks in South Korea during 2016-2022.
- In 2021-2022 winter, we detected 44 HPAI viruses from wild birds.

Methods

Sample collection, virus isolation and sequencing

- We collected samples from wild bird habitats in South Korea for routine AIV surveillance.
- We detected viruses using rRT-PCR and sequenced full-length genomes of the isolates using the Illumina MiSeq system.

Phylogeographic analysis

- For the phylogenetic analysis, time-scaled maximum clade credibility tree was constructed.
- To estimate the geographical viral transmission and the contribution of host species to viral spreading.
- We conducted Bayesian phylodynamic analysis and identified the statistical significance of each transition using Bayes Factor and posterior probability

Reassortment detection

- Whole-genome phylogenetic analysis was conducted to detect reassortment and define genotypes.

Results

- We isolated 43 H5N1 and one H5N8 viruses belong to clade 2.3.4.4b through wild bird surveillance (**Table 1**).
- Phylogenetic analysis reveals that 43 subtype H5N1 viruses form 2 different clusters; N1.G1 and N1.G2 (**Figure 1**).
- H5N8 isolate clustered with Asian H5N8 virus in 2020-2021 winter which indicating H5N8 viruses maintained more than one year (**Figure 1**).
- Phylogeographic phylodynamic analysis reveals that clade 2.3.4.4b spread worldwide through wild bird. (**Figure 2, 3**)
- The most frequent viral transition occurred from Europe to Central Asia and from wild birds to Domestic *Galliformes*. (**Figure 2, 3**)
- 2.3.4.4b HPAI viruses derived various genotype through reassortment with LPAI viruses. (**Figure 1, 4**)
- 43 H5N1 viruses divided into 5 distinct genotypes: N1.G1.1, N1.G2, N1.G2.1, N1.G2.2, N1.G2.2.1. (**Figure 5**)

Discussions

- Unlike the previous HPAIV outbreaks, clade 2.3.4.4b had maintained for 22 months, from May 2016 to March 2018 and, for 30 months from September 2020 to February 2023.
- Phylodynamic analysis showed the frequent spread from wild bird to domestic poultrys and the long-term circulation of the virus in wild birds.
- The H5N8 virus re-isolation indicates possible circulation of the virus in terrestrial wild birds for more than one year.
- These results indicate that the pattern of virus transmission through wild birds has been changing.
- Including this study, a wide variety of recombinant viruses having genes of LPAI viruses in wild birds have been reported.
- Viral reassortment can cause rapid viral biological characteristics changes such as mammalian infection.
- Thus, continuous monitoring on novel reassortant is necessary.
- As the virus is suspected to be well adapted in wild birds, and be enzootic, a response system should be prepared under the premise that wild birds are always infected.

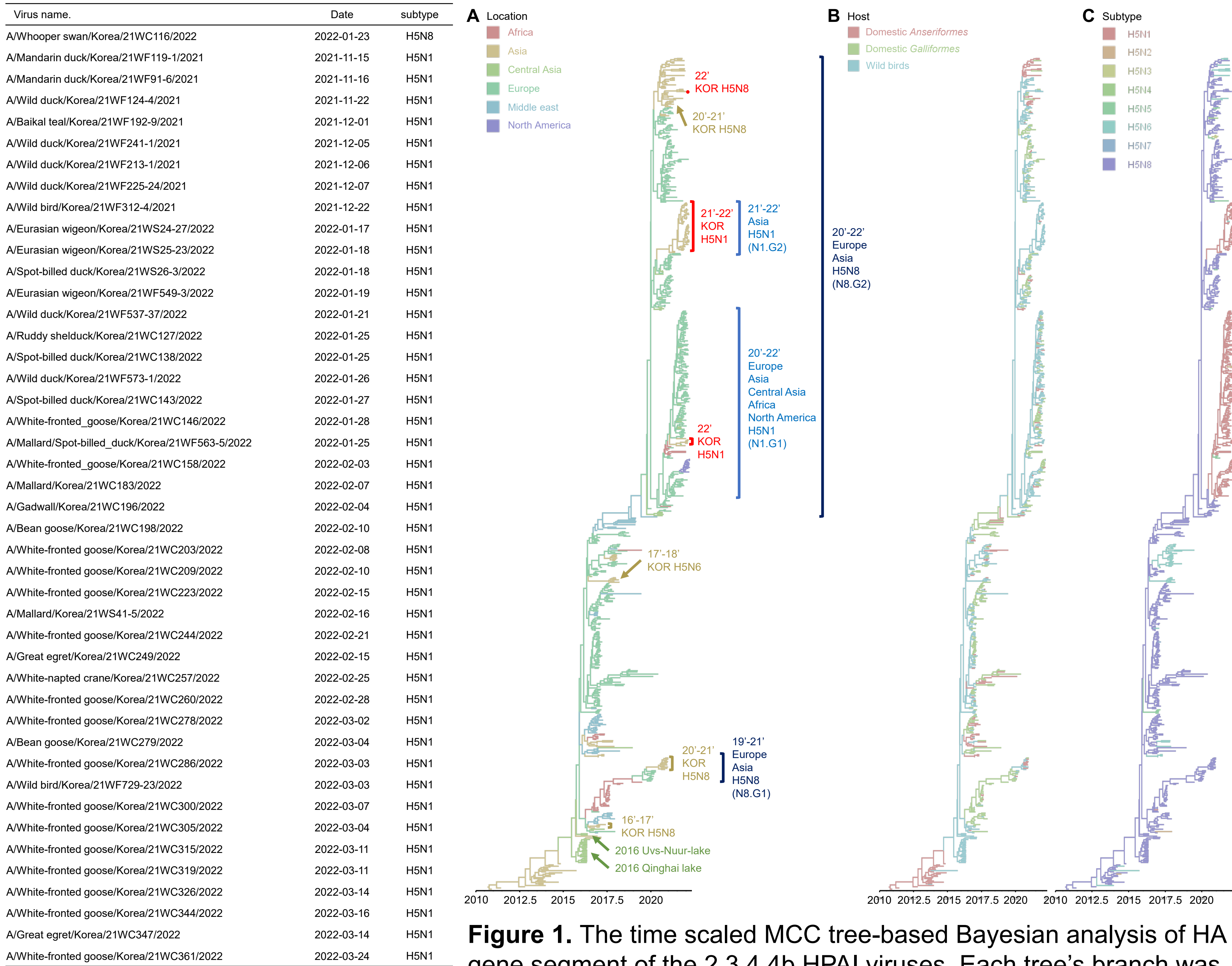


Table 1. Viruses isolated in this study

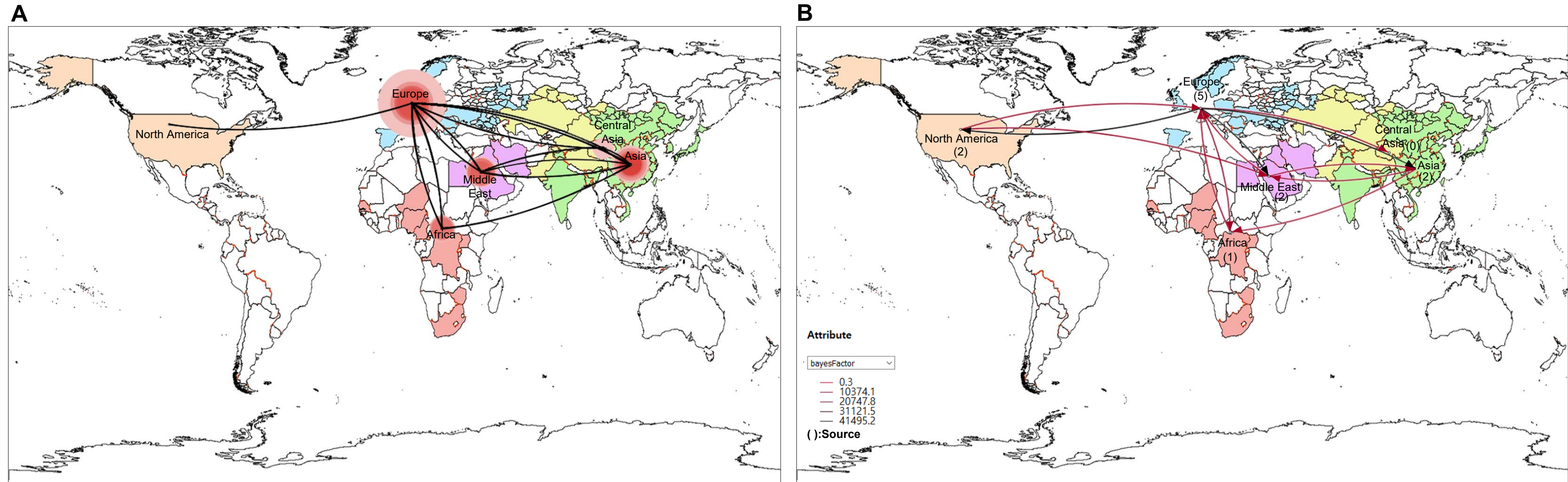


Figure 2. Visualization of phylogeographical transmission pathway on the global map.

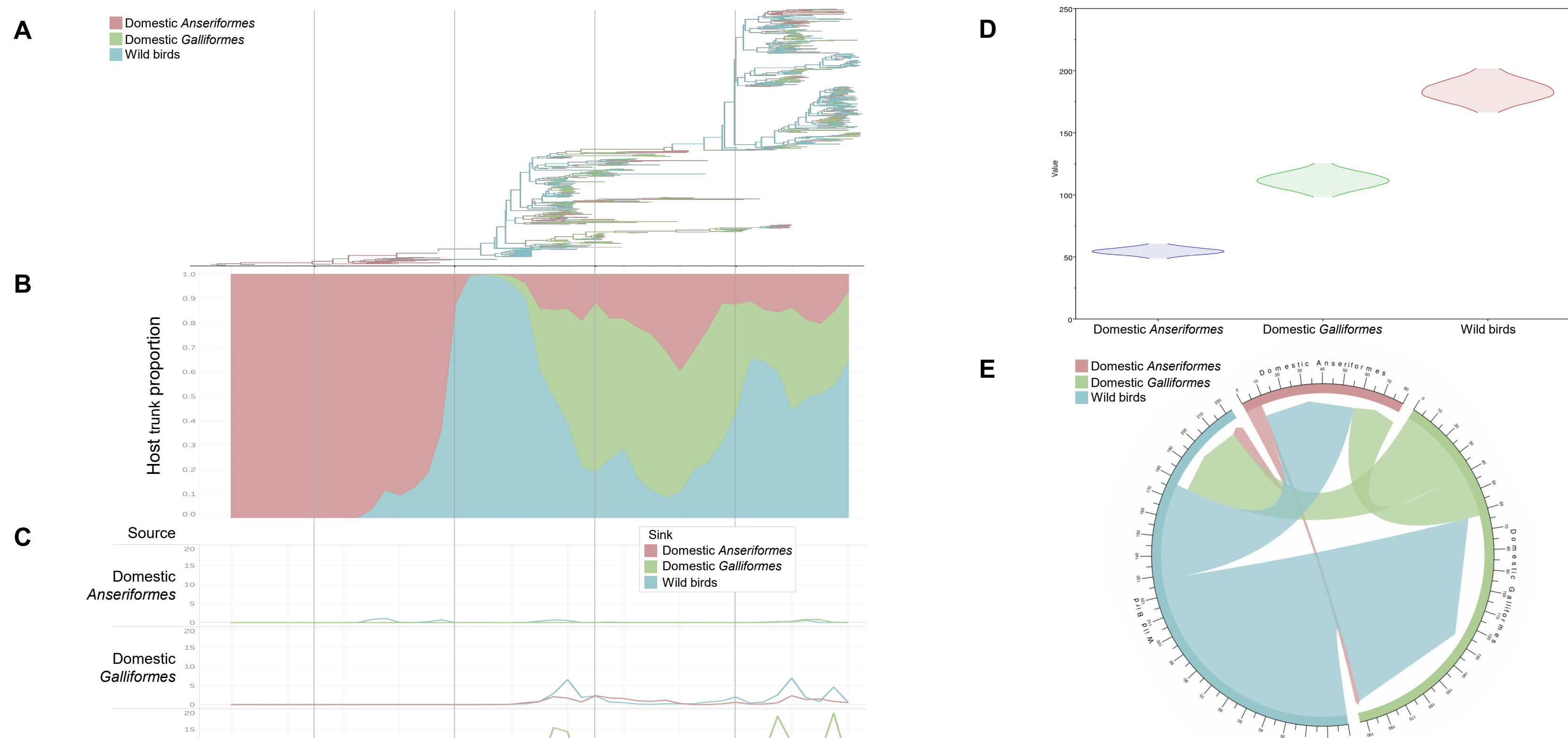


Figure 3. Phylogenetic estimation of viral transitions between host species.

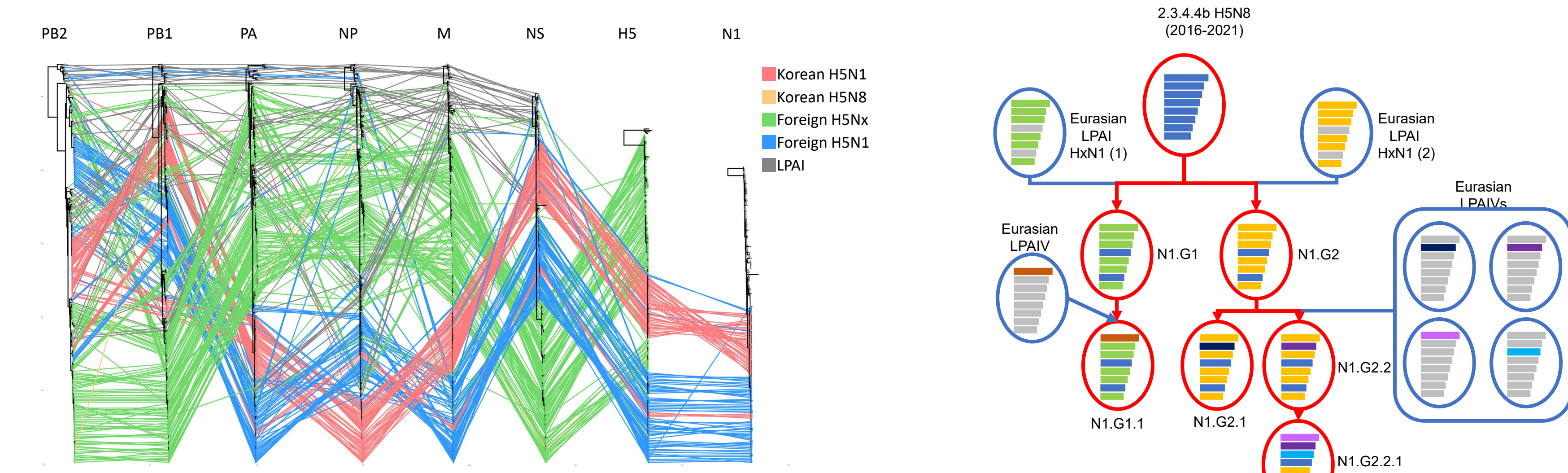


Figure 4. Tanglegram for each AIV gene segment. Connecting lines were colored according to the five groups.

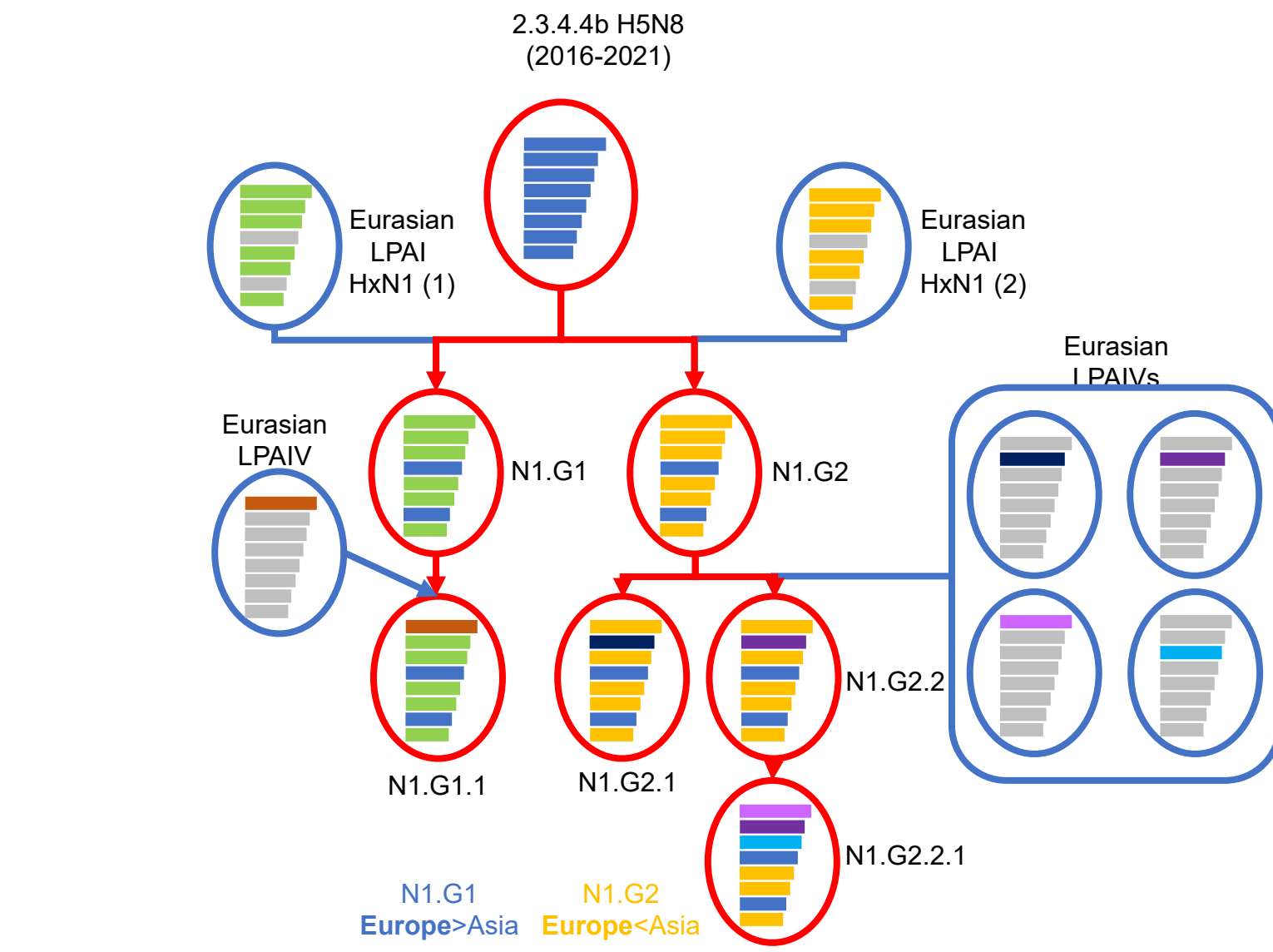


Figure 5. Reassortment dynamic of clade 2.3.4.4b. Each gene segment is represented by a bar of different length, from top to bottom: